Eureka! Solve this puzzle 4

Archimedes of Syracuse (287 – 212 BC) ran naked out into the streets crying 'Eureka!', excited after finding out the answer to King Hieron II's problem.

1 The following words relate to this topic about matter but are incomplete. Identify the missing letters and then use them and their matching numbers to complete the sentence below and discover the problem King Hieron had.

m a _ t e _ 1 6	a t o _ s 2	s _ b s t a n _ e 3 5	m o l _ _ u l e _ 4 5 8	_ a _ t i c _ e s 17 6 15
_ o l _ d 8 7	b _ n _ s 11 13	_ o r _ 9 2	_ r i t e r i _ 5 10	m _ t a _ s 4 15
l _ q u _ d 7 7	s _ l i d 11	g _ s 10	c o _ p o u n _ s 2 13	_ a t e r 12
s i l _ e _ 14 6	_ y d r _ _ e n 16 11 19	_ l e _ e _ t 4 2 18		

King Hieron II's problem

_ _ _ 13 7 13	_ _ _ 1 16 4	_ _ _ _ _ _ _ _ _ _ 19 11 15 13 8 2 7 1 16 8	_ _ _ 3 8 4	_ _ _ 10 15 15	_ _ _ 1 16 4	_ _ _ _ 19 11 15 13	_ _ 11 6	_ _ _ _ _ _ _ 6 4 17 15 10 5 4

_ _ _ _ 8 11 2 4	_ _ 11 9	_ _ 7 1	_ _ _ _ 12 7 1 16	_ _ _ _ _ _ 8 7 15 14 4 6	_ _ _ _ 12 16 4 18	_ _ k _ _ _ 2 10	_ _ _ 7 18 19	_ _ _ 1 16 4	_ _ _ _ _ _ 19 11 15 13 4 18	_ _ _ _ _ ? 5 6 11 12 18

2 Write each of the words listed above into a sentence about science.

3 **Research:** Use library resources to find out how Archimedes solved the king's problem. This should explain why he was naked at the time!

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